

CS3807 – Deep Learning Laboratory

Laboratory Guidelines and Project Evaluation Policy

1. General Laboratory Guidelines

1. The Deep Learning Laboratory is a **continuous evaluation** course. Students are expected to actively participate in every laboratory session.
2. Students must perform all laboratory experiments **independently**. Discussion of concepts is encouraged; however, code, reports and experimental results must be original.
3. **No coding or debugging support will be provided by the instructor.** The instructor will explain the underlying concepts, mathematical formulation, datasets, evaluation metrics and research methodology. Students are expected to independently design, implement, debug and validate their solutions.
4. Every experiment must be documented using **LaTeX**. Students are encouraged to prepare reports using **Overleaf**.
5. Source code for every experiment must be maintained in a **GitHub repository**. The repository should contain a README, source code, dataset information, dependency list and execution instructions.
6. Students are expected to follow good software engineering practices including modular programming, meaningful commit history, documentation and reproducible experiments.
7. Academic integrity must be maintained throughout the course. All project reports will be checked using **Turnitin**. Any form of plagiarism or unauthorized copying will be dealt with according to the University's academic regulations.

2. Semester Project Guidelines

1. Students may work individually or in teams of **up to three members**.
2. The instructor will provide **four project problem statements**. These topics will be allotted on a **first-come, first-served** basis.
3. After these topics are allotted, the remaining teams must select a project from a recent research paper published in a reputed Deep Learning journal or conference.
4. The same project topic will **not** be allotted to multiple teams.
5. The selected project should reproduce or extend an existing deep learning architecture and compare the obtained results with published work using appropriate evaluation metrics.
6. Students are encouraged to select papers published during the last five years.
7. Two formal project reviews will be conducted during the tutorial hours, followed by a final project demonstration and viva voce.

Recommended Publication Venues

Journals

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Multimedia (TMM)
- Pattern Recognition
- Neural Networks
- Neurocomputing
- Knowledge-Based Systems
- Expert Systems with Applications
- Applied Soft Computing
- Engineering Applications of Artificial Intelligence
- Signal Processing: Image Communication

Conferences

- CVPR, ICCV, ECCV
- NeurIPS, ICML, ICLR
- AAAI, IJCAI
- ACM Multimedia
- WACV

3. Project Evaluation Policy

Evaluation Component	Weightage
Problem Selection, Literature Survey and Review-I	10%
Methodology, Experimental Design and Review-II	20%
Implementation Quality, GitHub Repository, Code Documentation and Reproducibility	20%
Experimental Results, Performance Evaluation and Comparative Analysis with Existing Methods	20%
Final Demonstration and Technical Presentation	15%
Individual Viva Voce and Technical Understanding	15%
Total	100%

4. Competition-Based Project Evaluation

Students may alternatively participate in a Kaggle competition or any instructor-approved Deep Learning competition in place of a research-paper-based project. The competition should involve significant model development and experimentation.

Evaluation Component	Weightage
Competition Selection and Problem Understanding	10%
Data Preparation and Experimental Design	15%
Model Development, Hyperparameter Tuning and Innovation	25%
Leaderboard Performance and Comparative Analysis	20%
GitHub Repository, Documentation and Reproducibility	10%
Final Demonstration and Technical Presentation	10%
Individual Viva Voce and Technical Understanding	10%
Total	100%

Important Notes

- A higher leaderboard score alone does not guarantee higher marks.
- Greater emphasis will be placed on originality, implementation quality, reproducibility, experimental analysis and technical understanding.
- Marks may vary among team members based on individual contribution and viva voce performance.
- Every submission must include the GitHub repository link. Project reports must be prepared in LaTeX and submitted as PDF.